Research Paper Report

AboutMe: Using Self-Descriptions in Webpages to Document the Effects of English Pretraining Data Filters

Li Lucy et al., ACL 2024

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Question 1: Background/Context [Max 300 words]

Large language models are trained on massive web corpora, typically derived from Common Crawl. Before training, this raw data undergoes curation: automated filters remove text deemed low-quality, non-English, or toxic. These filters are presented as objective, performance-enhancing steps, but the specific decisions they encode about what counts as "quality" remain poorly documented and under-scrutinised.

This matters because pretraining data composition directly shapes model behaviour. Prior work has shown that what appears in training data influences what models can recall and generate (Kandpal et al., 2023; Razeghi et al., 2022). Gebru et al. (2021) argued for systematic dataset documentation, yet most curation pipelines still lack transparency about whose language they retain or discard.

Several strands of related work motivate this paper. Gururangan et al. (2022) demonstrated that GPT-3's quality classifier favours text from wealthier, more educated, and urban US school districts, providing early evidence that "quality" filtering encodes socioeconomic bias. Longpre et al. (2023) trained multiple LLMs on different data permutations and found that aggressive quality and toxicity filtering degrades downstream generalisation, revealing trade-offs in data pruning. On the language identification front, Caswell et al. (2020) and Kreutzer et al. (2022) documented systematic failures of langID tools on low-resource and code-switched text.

However, these studies were limited in scope: Gururangan et al. focused on a single filter and one demographic dimension (US school districts), while langID analyses lacked large-scale sociodemographic grounding. A fundamental obstacle has been the absence of metadata linking web text to its creators' social and geographic contexts, making it difficult to measure who is systematically excluded.

The AboutMe paper addresses this gap by constructing, for the first time, a large-scale dataset that pairs web content with its creators' self-reported identities, enabling systematic study of how ten different filters affect representation across social roles, topics, and geography.

Question 2: Contributions [Max 300 words]

The paper makes three main contributions: a novel dataset, a suite of extraction methods, and the first large-scale empirical study of how pretraining filters affect socially grounded web text.

The AboutMe dataset. The authors construct a corpus of 10.3 million websites from 24 Common Crawl snapshots (2020–2023), processed via CCNet. Each website is represented by two pages: an "About" page (identified through URL pattern matching) providing self-reported creator information, and a randomly sampled content page from the same hostname. This paired structure allows filter effects on general content to be traced back to creator demographics.

Social dimension extraction. The authors develop and apply methods to extract four dimensions from About pages. Topical interests are identified through TF-IDF vectorisation and balanced k-means clustering (k=50). Websites are classified as individuals or organisations using a random forest trained on pronoun ratios and named entity features (macro F1: 89.2). Social roles are extracted via a RoBERTa-based token classifier, domain-adapted through continued pretraining on About pages and fine-tuned on 1,000 annotated sentences (word-level F1: 0.898), with extracted roles mapped to the O*NET occupational taxonomy. Geographic affiliations are inferred using Mordecai3, a neural geoparser that disambiguates place names via contextual embeddings, tagging 63.2% of hostnames with country-level locations.

Filter evaluation. The study applies ten filters—four model-based quality classifiers (trained on Wikipedia, OpenWebText, and Wikipedia references), one perplexity scorer (KenLM trained on Wikipedia), Gopher's 19 rule-based heuristics, and four langID systems (fastText, CLD2, CLD3, langdetect)—to the annotated corpus. Through removal-rate analysis and OLS regression, the authors demonstrate that quality classifiers behave as implicit topical filters, penalising lower-prestige occupations, while langID systems disproportionately exclude English content from non-anglophone regions, particularly Asia.

Question 3: Critique [Max 300 words]

The paper's results largely support its conclusions, though with some caveats. The core claim—that filters impose implicit social preferences—is well-evidenced through consistent patterns across multiple filters, social dimensions, and analysis methods. The regression analysis controlling for topic, length, and geography strengthens the finding that langID bias against Asian regions persists independently of topical skew.

A key strength is the study's scale and grounding. By linking 10.3 million websites to self-reported creator identities, the authors move beyond anecdotal evidence to quantify filter effects at the population level. The paired page design—using About pages for metadata and separate content pages for filter evaluation—is methodologically clean and avoids conflating biographical text with general web content.

However, there are limitations. First, the study examines filters in isolation. In practice, curation pipelines layer multiple filters sequentially, and interaction effects between filters could amplify or mitigate the biases observed here. The authors acknowledge this but do not investigate it. Second, the dataset is restricted to websites with explicit About pages, which biases toward more professionally structured sites and may underrepresent casual creators whose content filters would most affect. Third, the occupational prestige analysis relies on O*NET salary data and Hughes et al.'s prestige ratings, both US-centric frameworks, yet the dataset spans global creators. Prestige norms vary across cultures, so these results may not generalise straightforwardly.

A broader limitation is that the study documents disparities in filter behaviour without directly measuring downstream model impact. Showing that filters remove more content from certain groups is necessary but not sufficient to demonstrate harm—the missing link is whether these removal patterns measurably degrade model performance for affected populations.

Despite these caveats, the paper makes a valuable empirical contribution to an understudied area and provides a reusable framework for future audits of data curation practices.

Question 4: Generative AI Usage [Max 100 words]

I used Claude (Anthropic) as a research aid throughout this assignment. Specifically, I used it to summarise and cross-reference sections of the paper and cited works for faster comprehension, to discuss and refine my understanding of the technical methods (balanced *k*-means, Mordecai3 geoparsing, perplexity filtering), and to review drafts of my answers for clarity and conciseness. All final answers reflect my own analysis and interpretation of the paper; Claude served as a sounding board rather than a content generator.